

Residual Analysis Transcript

Hello everyone, welcome to lesson 2 video 3. In this video, we will focus on residual analysis. Residuals are differences between observed and predicted values. Analysing them helps us to check the assumptions of regressions.

So far in our journey with linear regression, we have learned how to build a model. We have also learned how to find the best line. We have learned how to test whether the coefficient or the parameters of the model are significant.

But there is one final crucial step that we need to know, which is the assumptions of the model. Think of this as a health check for our model, right? A low p-value is great, but if the model fails to health check, or let's say if the model fails to health check, then our conclusion could be wrong. Today we are going to learn how to do the residual analysis to find out whether the assumptions are met or not, okay? So the idea is to check on our assumptions or the assumptions which linear regression model overlay.

The very first assumption that we have is the relationship between y and x has to be linear, okay? And the errors actually tells us the story. The errors tells us about every single assumption with linear regression model holds. If you think of, errors are nothing but the difference between the true value of y and the predicted value of y , right? So if we analyse the pattern of error, how are they distributed, how do they look like? Not only for one particular error, but for the entire data.

If we see their distribution and everything, we will be able to understand whether our assumptions are being met properly or not, okay? These are the key assumptions that we have in the model. First assumption, as I have told you, it's a linear assumption. The underlying relationship between x and y is linear, right? Then the second is independence, which is the error are independent of each other, okay? Then we have the normality assumption, which means that errors are normally distributed with mean 0 and equal or constant variance, which is also our last assumption, equal variance, right? We also call this as homoscedasticity, okay? When the variances are not equal, then that situation is called as heteroscedasticity, okay? If you think of this entire set of assumption, they are called as line assumptions.

And because they are line assumptions, I mean, this is also a parallel between we are fitting a line or a linear hyperplane in the data, right? In simple linear regression, it's a line because you have only two variables. So, we actually are dealing with line in the assumptions as well. Linearity, which is nothing but the relationship between x and y is linear, independence, the error terms has to be independent from each other, normality, the error terms has to be normally distributed with mean 0 and a constant error, which is homoscedastic or equal error, which is called as equal variance.

Now, the most important tool is to plot the residual plot. What do we see from residual plot generally and how does it generally looks like? This is what we are going to

understand. So, if we get to see some pattern in the residual plot, either increasing or there is some linearity or the data points are scattered in a random way, that's all good.

If there is some relationship, then it's not good. And that's what we are going to learn into this entire situation. Let's say here on the left hand side, we see that the residuals are all random.

There is no pattern, right? No obvious pattern. Hence, we are all good. There are more chances that the assumptions are met properly.

When you see a pattern, any type of pattern, in your residual curve, let's say, residual versus fitted value, it means that you have some problem there. A U-Sweep curve or the other curves indicates that there is a relationship between them. When there is a relationship, it also means that some assumptions of the model are not being met properly.

Similarly, when we check for Equal Variance or Homoscedastic Condition, remember the last assumption was of Homoscedasticity. If the errors are randomly distributed, say for example, the mean line is zero line here, we assume that the error has to be zero, right? And that's what our entire assumption is all around. So, if the errors are normally distributed with mean zero, everything should be randomly distributed above and below that mean line, right? But we should not have a pattern, let's say for example, here, if you go from left side to right hand side, you see a funnel looking shape, which means that the variance is not equally distributed, which means that there is a presence of Heteroscedasticity, okay? Then for to check for the normality, we check a normal Q-Q plot.

And if we get a normal Q-Q plot in that the probability values are low, then we mean that the errors are following normal distribution. Also, we check for Histogram of Residuals, so we first compute residuals as the difference between the true value and predicted value, and then we plot the histograms of them. If the histogram looks like a normally distributed curve, right, then in that case, we say that errors are normally distributed.

Let's see what does exactly that means. Let's say for example, you have the data here, and let's say we have used the same equation to predict the values. So let's say this is predicted, and we say is equals to 10 times of ads plus 90, okay? And you see, you have all these predictions here.

We can calculate errors, and you see errors as this minus, sorry, is equals to this minus or let's say it's a true value minus predicted value, okay? You got this, and you can extend it for all the variables, sorry, all the data points, and you'll get that entire data. Now, if errors are normally distributed, we can plot the histogram of plot for errors. Let's say insert, chart, you see that this is the error, a mean error, and you see an increasing pattern.

It means that something is systematically affecting the errors. The idea is to help you understand that once you have got the error values, you can actually create a histogram for it, okay? Once you create a histogram for the error values, if the histogram for error values looks normal, okay, it looks really normally distributed, and that's what we have to look for. If the entire error or the distribution of error, it looks like normally distributed, it means that they are following normal distribution.

But in this particular case, even this plot, the error plot that we have got here is also meaningful. If you see the errors are, the mean of errors is not going to be zero, right? If you take the mean of this entire data, sorry, I don't know if the mean is a function, average is, and you see that the averages are not closer to zero, right? Which means that the assumption that error follows normal distribution with mean zero and constant standard deviation, they are not being followed properly. That is where we have to investigate deeply what is the problem, and in that case, the assumption that we have made about the parameters of the model, like β_1 , could not be reliable, okay? So to fit the best possible model, you also would have to understand whether the assumptions are actually meeting well or not.

So that's what the diagnostic tools are. You have different types of diagnostic tools. You have residual versus fitted plot, you can actually plot residual versus fitted and then see for if there is a systematic pattern.

Let me show you that as well. So let's say you have this and you insert chart, okay? Now it is showing, now if you see here, this is error versus predicted value chart. You'll see that predicted value on the x-axis and, sorry, an error value on the y-axis.

you see that there is a relationship, there seems like a relationship, right? And that is what is not good. It should not have a relationship. We learned about it, if you remember.

So this one, right? So the fitted value and residual plot, it should not have a relationship. These are the cases when your data is not following the distribution or the assumption distribution that we have laid down, right? So the four assumptions that we have are linearity, independence, normality, and equivalence. So some of them might not be holding true in this case.

That's what it means. I hope this is all clear to everyone. I mean, there are cases as well when we might overlook some of these assumptions.

But those cases need to be very thoroughly investigated, okay? But at least from the point of view of understanding of the model, how to test for the model, you should be good with it now. In summary, residual analysis helps us to check whether the regression assumptions are satisfied by examining the patterns in residuals. We can identify potential outliers like nonlinearity, inequality, variances, outliers, and so on.

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